# Elementary Matrices



* **Lemma 1.1** Matrix multiplication is associative ie $A.(B.C) = (A.B).C$

* **Lemma 1.2** If $P = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$, $Q = \left[\begin{array}{c|c} A' & B' \\ \hline C' & D' \end{array}\right]$ are matrices s.t (i) $\exists R = P.Q$ and (ii) # columns of $A$ = # rows of $A'$

  then $R = \begin{bmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{bmatrix}$

* **Defn 1.3** A matrix $A$ is called invertible if $\exists\ B$ s.t $A.B = B.A = I$

* **Lemma 1.4** $A$ is invertible $\iff$ $A^T$ is invertible

* **Lemma 1.5** If $A, B$ and $C$ are matrices s.t $C = A.B$ and both $A$ and $B$ are invertible $\Longrightarrow$ $C$ is invertible

* **Lemma 1.6** Elementary Matrices $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} / \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}$

  → Effect of left and right multiplication

  → They are invertible

* **Defn 1.7** Reduced row echelon form (rref):

  ① The first non zero element in a row, ie the pivot, is 1. All the elements above and below a pivot are zero

  ② If row $i$ and row $j$ are two rows of rref s.t (i) $j > i$ and (ii) they both have pivot elements then row $j$ pivot lies to the right of row $i$ pivot

  ③ If row $i$ is all zeros then all the rows below it are also all zeros.

* **Lemma 1.8**

  $A$ (square matrix)

  ↙ invertible — non invertible ↘

  $\text{rref}(A) = I$ ; $\text{rref}(A) = \begin{bmatrix} P & Q \\ 0 & 0 \end{bmatrix}$ and also $\exists$ an invertible matrix $M$ s.t $\text{rref}(A) = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} M$

* **Theorem 1.9** If $A, B, C$ are square matrices s.t $C = A.B$

  $C$ is invertible $\iff$ Both $A$ and $B$ are invertible

  Hint Use 1.8 and 1.5

* Lemma 1.10 — A is an invertible matrix $\iff$ A is product of elementary matrices

* Defn 1.11 — Determinant : $\mathbb{R}^{n\times n} \longrightarrow \mathbb{R}$ s.t

① $\det(I) = 1$

② $\det\left(\begin{bmatrix} \vdots \\ a\rho + b\mathcal{Q} \\ \vdots \end{bmatrix}\right) = a\det\left(\begin{bmatrix} \vdots \\ \rho \\ \vdots \end{bmatrix}\right) + b\det\left(\begin{bmatrix} \vdots \\ \mathcal{Q} \\ \vdots \end{bmatrix}\right)$, ie det is linear in rows

③ If two rows of A are same then $\det(A) = 0$

* Lemma 1.12 — If E is an elementary matrix and A are square matrix then

$$\det(EA) = \det(E)\det(A)$$

* Theorem 1.13 — If A and B are square matrices of same size then

$$\det(AB) = \det(A)\det(B)$$ Hint 1.12, 1.10

* Lemma 1.14 — If A is a square matrix then $\det(A) = \det(A^T)$

* Theorem 1.15 — If $A, \rho, \mathcal{Q}$ are square matrices s.t $A = \begin{bmatrix} \rho & 0 \\ 0 & \mathcal{Q} \end{bmatrix}$ then

$$\det(A) = \det(\rho)\det(\mathcal{Q})$$

Hint (i) $\rho$ or $\mathcal{Q}$ is non invertible

(ii) $\rho$ and $\mathcal{Q}$ are invertible

$f(B) := \det\begin{bmatrix} B & 0 \\ 0 & \mathcal{Q} \end{bmatrix} \Big/ \det(\mathcal{Q})$

$f(B)$ satisfies 1.11 and $\exists$ a det map.

* Note 1.16 — Discussions from 1.1 to 1.15 can be extended to complex matrices.

* Defn 1.17 — $S_n$: Symmetric group: Group of permutations of n objects

Cyclic notation for elements of $S_n$

* Defn 1.18 — $GL_n$: General linear group

$$GL_n := \{ A \in \mathbb{R}^{n\times n} \text{ s.t } AB = I \text{ where } B \in \mathbb{R}^{n\times n} \}$$

* Defn 1.19 $SL_n$ : Special linear group

$$SL_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } \det(A) = 1 \}$$

* Defn 1.20 $O_n$ : Orthonormal group

$$O_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } A \cdot A^T = A \cdot A = I \}$$

* Defn 1.21 $SO_n$ : Special Orthonormal group

$$SO_n := \{ A \in \mathbb{R}^{n \times n} \text{ s.t } A \cdot A^T = I \text{ and } \det(A) = 1 \}$$

* Lemma 1.22 Let A and B are matrices and $x$ a column vector s.t.

(i) $Ax = 0$

(ii) $BAx = 0$

(a) $\text{rank}(B) = \#\text{columns of } B \implies \dim(\text{solution space of (i)}) = \dim(\text{solution space of (ii)})$

(b) $\text{rank}(B) < \#\text{columns of } B \implies \dim(\text{solution space of (i)}) \leq \dim(\text{solution space of (ii)})$

# Groups

* **Defn 2.1** Law of composition on a set $S$ is a function $f : \underbrace{S \times S}_{\text{cartesian product}} \to S$

* **Def 2.2** $e \in S$ is called identity w.r.t a law of composition on set $S$ if $\forall a \in S \quad e.a = a.e$.
Caution: A set might not contain an identity element w.r.t a law of composition

* **Defn 2.3** A transposition is an element of $S_n$ that switches only two objects.

* **Defn 2.4** Group: A set $G$ together with a law of composition $(.)$ s.t
① $(a.b).c = a.(b.c) \quad \forall a, b, c \in G$
② $\exists e \in G$ s.t $ea = a.e \quad \forall a \in G$
③ $a \in G \Rightarrow \exists a^{-1} \in G$ s.t $a.a^{-1} = a^{-1}.a = e$

* **Example 2.5** $S_3$ : Symmetry group of 3 elements
$x = (123) \quad y = (12)$ generators
$x^3 = 1 \quad y^2 = 1 \quad yx = x^2y$ defining relations
$S_3 = \{1, x, x^2, y, xy, x^2y\}$

* **Defn 2.6** Subgroup: A non empty subset $H \subset G$ s.t
① if $a, b \in H$ and $c = a.b \Rightarrow c \in H$
② if $a \in H \Rightarrow a^{-1} \in H$

* **Example 2.7** $SO_n \subset SL_n \subset GL_n$
$C^\infty$

* **Defn 2.8** $H \subset G$ is called normal subgroup, ie $H \trianglelefteq G$
if $\forall g \in G$ and $h \in H \quad ghg^{-1} \in H$

* **Defn 2.9** Order of a group $G$ is the number of unique elements it has.

* Defn 2.10 Let $U$ be the subset of group $G$. Then the group generated by $U$ is defined as the smallest subgroup $H$ of $G$ that contains all elements of $U$

* Defn 2.11 Cyclic subgroup: The group generated by a single element $x$ of a group

$$\langle x \rangle := \{ \ldots, x^{-2}, x^{-1}, e, x, x^2 \ldots \}$$

Order of $x :=$ Order of $\langle x \rangle$

* Defn 2.12 Group Homomorphism: $\varphi : G \longrightarrow G'$ s.t

$$\varphi(a.b) = \varphi(a).\varphi(b) \quad \forall\, a, b \in G$$

Image of $\varphi := \{ \varphi(g) : g \in G \}$

Kernel of $\varphi := \{ g \in G : \varphi(g) = e' \}$

* Lemma 2.13 (i) $\varphi(e) = e'$

(ii) $\varphi(a^{-1}) = \varphi(a)^{-1}$

(iii) if $H < G \implies \varphi(H) < G'$

* Lemma 2.14 (i) Kernel is a normal subgroup of $G$

(ii) Image is a subgroup of $G'$

* Lemma 2.15 A homomorphism $\varphi$ is injective i.e one to one if Kernel of $\varphi$ if $\{e\}$

* Defn 2.16 Isomorphism: A bijective group homomorphism

* Defn 2.17 Automorphism: An isomorphism from group to itself

* Defn 2.18 $a, b \in G$ are called conjugates if $\exists\, g \in G$ s.t $b = gag^{-1}$

* Defn 2.19 Let $H \subset G$ and $a \in G$ then

Left coset of $a := aH = \{a.h \text{ where } h \in H\}$

↳ All the left cosets wrt a subgroup have the same size

* Theorem 2.20 Let $H \subset G$ then $[G] = [H][G:H]$

order of $G$ | Order of $H$ | # of left cosets

Hint: $a R b$ (related) if $\exists h \in H$ s.t $a = b.h$ is an equivalence relation ⇓

$[G] = [H][G:H]$ ⇐ Left cosets are non intersecting (2.19)

* Corollary 2.21 Let $\varphi : G \to G'$ be a group homomorphism,

$[G] = [\text{Kernel}]\,[\text{Image}]$

* Correspondence Theorem 2.22: Let $\varphi : G \to G'$ be a surjective group homomorphism

(i) $[H \subset G \text{ s.t } K \subset H] \xrightarrow[\text{bijective map}]{f} [\text{Subgroups of } G']$

(ii) If $\{H \lhd G \text{ and } K \subset H\} \xrightarrow[\text{bijective map}]{g} \{H' \lhd G\}$

* Hint: (a) We need $K \subset H$ (i) and (ii) for $f$ and $g$ to be injective

(b) we need $\varphi$ to be surjective for (I) $f, g$ to be onto

(II) $\varphi(H) \lhd G'$ where $H \lhd G$ and $K \subset H$

* Defn 2.23 Quotient Group: Let $H \lhd G$, $\bar{a} := aH$ left coset of $a \in G$ then

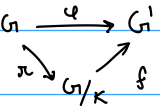
$G/H := \{\bar{a} : a \in G\}$ together with the law of composition $\bar{a}.\bar{b} = \overline{a.b}$ form a group

Further $\varphi : G \to G/H$ where $\varphi(a) = \bar{a}$ is a group homomorphism

Hint: Prove that the law of composition is independent of the choice of representative element and therefore valid. Then this law carries over the group structure.

* First Isomorphism Theorem 2.24

Let $\varphi: G \to G'$ be a surjective homomorphism and $K$ be its kernel, then $f: G/K \to G'$ defined by $f(\bar{a}) := \varphi(a)$ is a group isomorphism.




* Lemma 2.25

If $W_1$ and $W_2$ are subgroups of $G$, then the following are also subgroups

$$W_1 \cap W_2 := \{ g \in G : g \in W_1 \text{ and } g \in W_2 \}$$

$$W_1 + W_2 := \{ g \in G : g = a.b \text{ or } g = b.a \text{ where } a, b \in W_1 \cup W_2 \}$$

* Usually $W_1 + W_2 \neq W_1 \cup W_2$

* Lemma 2.26

All the subgroups of $\mathbb{Z}^+$ (additive group of integers) have the form $n\mathbb{Z}$ where $n \in \mathbb{Z}$.

Hint: Use well ordering property of $\mathbb{Z}$

* GCD Defn 2.27

If $a, b \in \mathbb{Z}$, then $d \in \mathbb{Z}$ is called gcd of $a$ and $b$ if

(i) $d > 0$

(ii) $d$ divides $a$ and $b$

(iii) if $c \in \mathbb{Z}$ divides $a$ and $b$ then $c$ divides $d$

* Lemma 2.28 If $a, b, d \in \mathbb{Z}$ then $d = g.c.d(a, b) \implies d\mathbb{Z} = a\mathbb{Z} + b\mathbb{Z}$

* Defn 2.29 $a, b \in \mathbb{Z}$ are called relatively prime if $g.c.d(a, b) = 1$

* Defn 2.30 Let $p \in \mathbb{Z}$ s.t if $p$ divides $a.b \implies p$ divides either $a$ or $b$

* Lemma 2.31 $p \in \mathbb{Z}$ is prime $\iff \nexists\ n \in \mathbb{Z}$ s.t $1 < n < p$ and $n$ divides $p$.

* L.C.M

defn 2.32 If $a, b \in \mathbb{Z}$, then $l \in \mathbb{Z}$ is called l.c.m of a and b if

(i) $l > 0$

(ii) a and b divides $l$

(iii) if a and b divides $c \in \mathbb{Z}$ then $l$ divides $c$

* Lemma 2.33 If $a, b, l \in \mathbb{Z}$ then $l = l.c.m(a,b) \iff l\mathbb{Z} = a\mathbb{Z} \cap b\mathbb{Z}$

* Lemma 2.34 $g.c.d(a,b) \times l.c.m(a,b) = a.b$

Proof (i) $a.b/d$ is divisible by $l$ using 2.32

(ii) $l.d$ is divisible by $a.b$ using $d = ma + nb$

* Defn 2.48 Modular arithmetic $f: \mathbb{Z}^+ \to \mathbb{Z}/n\mathbb{Z}$ where $n \in \mathbb{Z}$ is a group homomorphism (using 2.23)

further (i) $f(a+b) = f(a) + f(b)$

(ii) $f(a.b) = f(a).f(b)$

(iii) $f(a)(f(b) + f(c)) = f(a)f(b) + f(a)f(c)$

Congruence Relation $a \equiv b$ if $a-b$ is divisible by $n$

# Vector Space

* Defn 3.1 A field $F$ is a set together with two laws of composition, ie addition $(+)$ and multiplication $(\cdot)$ s.t

(i) $(F,+)$ is an abelian group with identity element denoted by $0$

(ii) $(F^*,\cdot)$ is an abelian group, where $F^* = \{F - \{0\}\}$. It's identity element is denoted by $1$.

(iii) $\forall\ a,b,c \in F \qquad a.(b+c) = ab + ac$

* lemma 3.2 $F_p = \mathbb{Z}/p\mathbb{Z}$ where $p \in \mathbb{Z}$ is prime is an field.

Hint: Proof by contradiction. Let $p = a.b$ with $1 < a,b < n$

show that closure fails for $F_p^*$

* lemma 3.3 $F_p^*$ is an group of order $p-1$ and is cyclic, ie

$$F_p^* = \{1, \alpha, \alpha^2 \ldots, \alpha^{p-2}\} = \{\bar{1}, \bar{2}, \ldots, \overline{p-1}\}$$

The generator $\alpha$ of $F_p^*$ is called primitive root modulo $p$.

* Defn 3.4 The <u>characteristic of a field</u> is the order of $\langle 1 \rangle \subset (F,+)$

* lemma 3.5 <u>characteristic of a field</u>

- Finite → It is a prime number
- Infinite → Then it is called zero

Hint: Same as 3.2

* Defn 3.6 A vectorspace $V$ over a field $F$ is a set together two laws of composition, ie. vector addition and scalar multiplication s.t

(i) $V$ is an abelian group with scalar addition. It's identity element is denoted by $0_V$

(ii) if $a \in F$, $v \in V$ then $a.v \in V$

(iii) if $v \in V$ $\quad 1.v = v$

(iv) if $v, w \in V$ and $a, b \in F \longrightarrow$

$$a(v+w) = av + aw$$

$$(a+b)v = av + bv$$

$$a.(b.v) = (a.b)v$$

* Lemma 3.7 Let $v \in V$, $0_F \in F \implies 0_F.v = 0_V$

* Defn 3.8 A subspace of a vectorspace $V$ over field $F$ is a non empty subset s.t
① if $v, w \in V$ then $v + w \in V$
② if $a \in F$, $v \in V$ then $a.v \in V$

* Lemma 3.9 If $W_1$ and $W_2$ are subspaces of $V$ then
① $W_1 \cap W_2$ is a subspace of $V$
② $W_1 + W_2$ is a subspace of $V$, where $W_1 + W_2$ is the group generated by $W_1 \cup W_2$ (2.25)

* Defn 3.10 Ordered Set → Order of elements matter
→ Repetition of elements is not allowed.
{Note: It is different from real analysis}

* Defn 3.11 The Span of an ordered set $S = \{v_1, v_2 \ldots\}$ with $v_i \in V$ is defined as the subspace generated by the linear combinations of $v_i$'s, i.e.

$$\text{Span}(S) := \left\{ \sum_i a_i v_i \text{ where } a_i \in F, v_i \in V \right\}$$

* Defn 3.12 A vector space $V$ is called finite dimensional if $\exists$ a finite set $S$ s.t
$$V = \text{Span}(S)$$

* Defn 3.13 An ordered set $S = \{v_1, v_2 \ldots, v_n\}$ with $v_i \in V$ is called linearly independent if
$$a_1 v_1 + a_2 v_2 \ldots + a_n v_n = 0_V \implies a_i = 0 \text{ where } 1 \le i \le n$$

* Defn 3.14 An empty set $\phi$ is linearly independent
$\text{Span}(\phi) = \{0_V\}$

* Defn 3.15 A basis $[B]$ of vectorspace $V$ is an subset of the vectors of $V$ s.t
① $[B]$ is linearly independent
② $\text{Span}(B) = V$

* Lemma 3.16 Let $L$ be an linearly independent set of finite dimensional $V$, $S$ be an set of $V$ s.t $Span(S)=V$ then $\dim(Span(L)) \leq \dim(Span(S))$

Hint: Proof by contradiction $\dim(Span(L)) > \dim((Span S))$

* Defn 3.17 The dimension of a finite dimensional vectorspace $V$ is defined as the number of vectors in its basis

* Lemma 3.18 Let $V$ be a finite dimensional vectorspace and, $[\mathcal{B}]$ and $[\mathcal{B}]'$ are its two basis associated by $[\mathcal{B}]' = [\mathcal{B}]P \Longrightarrow$ $P$ is invertible
$P$: base change matrix

If $v \in V$ then $v = [\mathcal{B}]'x' = [\mathcal{B}]x$ ($x$: coordinate vector of $v$ w.r.t $[\mathcal{B}]$)

$x = Px'$

* Defn 3.19 The two subspaces $U_1$ and $U_2$ of $V$ are called linearly independent if

$$U_1 \cap U_2 = \{0_v\}$$

* Defn 3.20 A vectorspace $V$ is called direct sum of subspace $U_1, U_2 \dots, U_k$ if

① $U_1, U_2 \dots, U_k$ are linearly independent

② $V = U_1 + U_2 + \dots U_k$

It is denoted by $V = U_1 \oplus U_2 \oplus \dots \oplus U_k$

* Lemma 3.21 If $W_1$ and $W_2$ are subspace of $V$ then

$$\dim(W_1) + \dim(W_2) = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$$

# Linear Transformation



* **Defn 4.1** Let $V, W$ be vectorspaces over the same field $F$. Then a map $T: V \to W$ is called a linear transformation if

  ① $T(v_1 + v_2) = T(v_1) + T(v_2)$ $\quad \forall v_1, v_2 \in V$

  ② $T(av) = aT(v)$ $\quad \forall v \in V, \forall a \in F$

* **Defn 4.2**

  Kernel of $T := \{ v \in V : T(v) = 0_W \}$

  Image of $T := \{ T(v) : v \in V \}$

* **Lemma 4.3**

  Kernel is a subspace of $V$

  Image is a subspace of $W$

* **Lemma 4.4**

  $\dim(\text{Kernel}) + \dim(\text{Image}) = \dim(V)$

  nullity $\quad$ rank

  Hint $[B] := \{ \underbrace{a_1, a_2, \ldots, a_k}_{\text{Basis of Kernel}}, \underbrace{b_1, \ldots, b_m}_{\text{Basis of } V} \}$

  Show $\{T(b_1), T(b_2), \ldots, T(b_m)\}$ is a basis of Image

* **Lemma 4.5**

  Let $T: V \to W$ be a linear transformation, $P$ a subspace of $V$

  then $T(P)$ is an subspace of the Image

* **Defn 4.6** Let $T: V \to W$ be a linear transformation, $[B], [C]$ be the basis of $V$ and $W$ respectively

  Then the matrix of linear transformation $T$ wrt $[B]$ and $[C]$ is defined as a matrix $A$ s.t

  $$T([B]) = [C]A$$

**Lemma** If $x, y$ be the coordinate vectors of $v \in V$ and $T(v) \in W$ wrt $[B]$ and $[C]$ respectively

then $Y = Ax$

**Lemma** If $[B']$ and $[C']$ are another sets of basis of $V$ and $W$ s.t

$$[B'] = [B]P \qquad [C'] = [C]Q$$

then $A'$, the matrix of linear transformation $T$ wrt $[B']$ and $[C']$ is given by

$$A' = Q^{-1} A P$$

* Defn 4.7 A linear transformation $T: V \longrightarrow W$ is called an Isomorphism if $T$ is bijective.

Example: A $n$ dim vectorspace $V$ over field $F$ is isomorphic to the vectorspace $F^n$.

* Lemma 4.8 Let $A_{n\times n}$ be an square matrix then

$AX = 0 \iff A$ is invertible

if and only if $x = 0$

Proof (i) $A$ is invertible $\Rightarrow$ let $AX = 0 \Rightarrow A^{-1}AX = A^{-1}0 \Rightarrow x = 0$

(ii) Let $AX = 0$ only for $x = 0 \Rightarrow$ columns of $A$ are linearly independent

$[A]$ is an basis of $\mathbb{R}^n$, $[I]$ is another basis of $\mathbb{R}^n \therefore \exists P$ s.t. $[A]\underbrace{P}_{\text{invertible}} = [I]$

* Lemma 4.9 Let $A_{m\times n}$ be a matrix and $E_{m\times m}$ be an invertible matrix, then

$\dim(\text{colSpace}(A)) = \dim(\text{Column Space}(EA))$

Hint use 4.4, 4.5

* Lemma 4.10 Let $A_{m\times n}$ be a matrix then

$\dim(\text{colSpace}(A)) = \dim(\text{RowSpace}(A))$

Hint Use 1.8, 4.9

* Defn 4.11 A linear operator $T$ is an linear transformation from a vectorspace $V$ to itself.

- Invertible / $T$ is bijective / kernel $= \{0_v\}$ / Image $= V$ → Transformation matrix is invertible
- Singular → Transformation matrix is non invertible

* Lemma 4.12 Let $T: V \to V$ be a linear operator, then the following conditions are equivalent

(i) $V = \text{kernel}(T) \oplus \text{Image}(T)$

(ii) $\text{kernel}(T) \cap \text{Image}(T) = \{0_v\}$

(iii) $\text{kernel}(T) + \text{Image}(T) = V$

These don't mean $T$ is invertible

Eg: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Hint: Use 4.4, 3.21

* Lemma 4.13 Let $T:V \longrightarrow V$ be a linear operator, $[B], [B']$ be two different basis of $V$ associated by $[B'] = [B]P$ and $A, A'$ be the transformation matrix wrt $[B]$ and $[B']$ then

$$A' = P^{-1}AP, \text{ further}$$

$A$ and $A'$ are called similar matrices as they represent the same linear transformation

* Defn 4.14 Let $T:V \longrightarrow V$ be a linear operator, then a subspace $W$ of $V$ is called T invariant if $T(W) \subset W$

* Lemma 4.15 Let $T:V \longrightarrow V$ be an linear operator, $W$ be an T-invariant subspace of $V$, $\{w_1, w_2 \ldots, w_k\}$ be an basis of $W$, $[B] := \{w_1, w_2 \ldots, w_k, v_1, v_2 \ldots, v_n\}$ be a basis of $V$, $f: V \longrightarrow V/W$ be a map defined by $f(v+W) := \bar{v}$

(i) $\bar{v}_1 + \bar{v}_2 = \overline{v_1 + v_2}$

(ii) $a\bar{v}_1 = \overline{av_1}$

} Similar to 2.23

(iii) $V/W$ along with (i) and (ii) forms a vectorspace

(iv) $f: V \longrightarrow V/W$ is a linear transformation

(v) $[\bar{B}] := [\bar{v}_1, \bar{v}_2 \ldots, \bar{v}_n]$ is an basis of $V/W$.

(vi) $\varphi: V/W \longrightarrow V/W$ given by $\varphi(\bar{v}) := f(T(v))$ is an linear operator

(vii) The transformation matrix of $T$ wrt $[B]$ has the form $A = \begin{bmatrix} P_{k\times k} & Q_{k\times n} \\ O_{n\times k} & R_{n\times n} \end{bmatrix}$

(vii)(a) The transformation matrix of $\varphi$ wrt $[\bar{B}]$ is $R_{n\times n}$

* Defn 4.16 Eigen vector Let $V$ be a vectorspace over field $F$, $T:V \longrightarrow V$ a linear operator. Then $v \in V$ is called an eigen vector of $T$ with eigen value $\lambda$ if (i) $v \neq 0_v$

(ii) $T(v) = \lambda v$ where $\lambda \in F$

* Lemma 4.17 An eigen vector is an one dimensional T-invariant subspace.

* Lemma 4.18 Let $V$ be a vectorspace over field $F$, $T:V \longrightarrow V$ a linear operator, $\lambda \in F$ then

$\lambda I - T : V \longrightarrow V$ is a linear operator

* Lemma 4.19 Let $v \in V$ be an eigen vector of $T: V \to V$ with eigen value $\lambda$, then

characteristic polynomial of $T$ $\{ \det(\lambda I - A) = 0$ where $A$ is the transformation matrix of $T$ $\Longleftarrow$ $\lambda I - T$ is singular $\Longleftarrow$ $v \neq 0$, is in the null space of $\lambda I - T$

* Lemma 4.20
(i) The characteristic polynomial of $T: V \to V$ is independent of the choice of basis
(ii) Similar matrices have the same characteristic polynomial
(iii) The co-efficients of characteristic polynomial is independent of the choice of basis
(iv) Eigen values are roots of the characteristic polynomial

* Lemma 4.21 Eigen vectors corresponding to distinct eigen values are linearly independent.

Hint: Proof by induction.

* Lemma 4.22 If $T: V \to \overset{n\dim}{V}$ has $n$ distinct eigen values then $\exists$ basis $[\mathcal{B}]$ of $V$ s.t the transformation matrix of $T$ w.r.t $[\mathcal{B}]$ has the diagonal form, ie. $\begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix}$

* Schur's Lemma 4.23 Let $T: V \to \overset{n \text{ dim V.S over } F}{V}$ be a linear operator,

If the characteristic polynomial $\Longrightarrow$ $\exists$ a basis $[\mathcal{B}]$ of $V$ s.t the transformation of $T$ factors into linear terms matrix of $T$ w.r.t $[\mathcal{B}]$ is upper diagonal $\begin{bmatrix} \lambda & * \\ 0 & R \end{bmatrix}$

Hint Use 4.20(iv), 4.17, 4.15 (vi), (viii) to prove upper diagonal

Proof: That the transformation is not always diagonalizable :-

Let $v_1$ be the eigen vector of $T: V \to V$, $V_1$ the span of $v_1$,

$f: V \to V/V_1$ where $f(v_2 v_1) = \bar{v}$

Let $\bar{v}_2$ be the eigen vector of $\varphi: V/V_1 \to V/V_1$ $\varphi(\bar{v}) := f(T(v))$

$v_2' = v_2 + c v_1$ $T(v_2) = \lambda_2 v_2 + n v_1$

$T(v_2') = T(v_2) + T(c v_1) = \lambda_2(v_2' - c v_1) + c\lambda_1 v_1 = \lambda_2 v_2' + (n + c(\lambda_1 - \lambda_2)) v_1$

then if $\lambda_1 \neq \lambda_2$ we can diagonalize

if $n = 0$ we can diagonalize

if $\lambda_1 = \lambda_2$ and $n \neq 0$ we cannot diagonalize

* Defn 4.24

Let $V$ be a vectorspace over field $F$ and $T: V \to V$ a linear operator. Then $x \in V$ is called a <u>generalized eigen vector</u> of $T$ with eigen value $\lambda$ and exponent $d$ if

① $x \neq 0_V$ ② $(T-\lambda I)^d x = 0$

③ If $(T-\lambda I)^k x = 0 \implies k \geq d$

* Lemma 4.25

Let $x \in V$ be a generalized eigen vector of $T: V \to V$ with eigen value $\lambda$ and exponent $d$,

$$u_j := (T-\lambda I)^j x \quad \text{with} \quad 0 \leq j \leq d-1$$

$$[B] := \{u_1, u_2, \ldots, u_{d-1}\}$$

$$X := \text{Span}([B])$$

then

(i) $B$ is linearly independent

(ii) $X$ is $T$ invariant

(iii) $T([B])$ has the form $\begin{bmatrix} \lambda & & 0 \\ 1 & \lambda & \\ & \ddots & \ddots \\ 0 & & 1 \; \lambda \end{bmatrix}_{d \times d}$

Proof Hint (i) Show that any non trivial combination $[B]$ is an generenerelized eigenvector $\therefore \neq 0_V$

* Lemma 4.26

A generalized eigen vector $x$ of $T: V \to V$ has unique eigen value $\lambda$ and exponent $d$

Proof Hint: Use binomial expansion

$$(x+y)^n = \sum_{k=0}^{n} {}^nC_k \, x^k y^{n-k}$$

* Jordan Decomposition Theorem 4.26

Let $T$ be a linear operator on a finite dimensional complex vectorspace $V$. Then $\exists$ a basis $[B]$ of $V$ s.t the transformation matrix of $T$ w.r.t $[B]$ has the Jordan form.

Proof: We will prove ①-③. They together lead to the basis $[B]$

① $V = V_1 \oplus V_2 \oplus \cdots \oplus V_r$

② $V_i = \text{Span}([u_{i,0}, u_{i,1}, \ldots, u_{i,d_i-1}])$, $1 \leq i \leq r$

③ $u_{i,j} = (T-\lambda_i) u_{i,j-1}$, $1 \leq j \leq d_i - 1$

generalized eigen vector with eigen value $\lambda_i$ and index $d_i$

We know that ①, ②, ③ are true for 1 dim V.

We will use principle of induction, ie assuming the statement is true for complex vector space of dimensions 1 to n-1 prove it for n dim complex vectorspace.

* Let $T: V \to V$ be a linear operator on n dim complex vector space V. Since its characteristic polynomial factors into linear terms $\Longrightarrow$ $\exists$ an eigen vector $v$ with eigen value $\lambda$.

If w.r.t basis [B] T has jordan form then so does $T-\lambda I$. Therefore for simplicity we we work with $T+\lambda I$ and refer to it as T, then

$T: V \to V$ is an linear operator and its nullspace is non-trivial.

Let $K_i$ be the kernel and $U_i$ be the image of $T^i: V \to V$. Since V is finite dimensional let $m \in \mathbb{N}$ be the smallest number s.t

$$K_i = K_j \quad \text{and} \quad U_i = U_j \qquad \forall\, i, j \geq m$$

then let $v \in K_m \cap U_m$, since $T^m v = 0$ and $\exists\, x \in V$ s.t $T^m x = v$

$T^{2m} x = 0 \Rightarrow x \in K^m \;\therefore\; v = T^m x = 0$

Using 4.12 $V = K_m \oplus V_m$

$\dim(V_m) < n$ $\therefore$ $\exists$ jordan generators. If $\dim(K_m) < n$ the proof is complete otherwise T is an idempotent operator

* $T: V \to V$ be an idempotent eigen vector, prove $\exists$ a basis [B] s.t T has jordan form.

Let K be the kernel of T and U be the image of T. Since $\dim(U) < n$ $\exists$ jordan generators for it ie. $U = U_1 \oplus U_2 \cdots \oplus U_r$ and

$$U_i = \text{Span}([u_{i,0}, u_{i,1} \cdots u_{i,d_i-1}]), \qquad 1 \leq i \leq r,$$

$$u_{i,j} = (T)^j v_i, \qquad 1 \leq j \leq d_i - 1$$

generalized eigen vector with eigen value 0 and index $d_i$

Let $w_i \in V$ s.t $T w_i = v_i$ then $w_i$ is an generalized eigenvector with eigen value 0 and index $d_i + 1$

Let $W_i = \text{Span}([u_{i,0}, u_{i,1} \cdots u_{i,d_i}])$ where $u_{i,j} = T^j w_i$

$W = W_1 + W_2 + \cdots + W_r$

Let $w \in W_n \cap W_k$ then $Tw \in V_n \cap V_k$ since $V_n \cap V_k = 0$

$w \in K$ Since $W_n \cap K = \text{Span}(u_{n,d_n}) = \text{Span}(v_{n,d_n-1})$

and $W_k \cap K = \text{Span}(u_{k,d_k}) = \text{Span}(v_{k,d_k-1})$

since $v_{n,d_n-1}$ and $v_{k,d_k-1}$ are linearly independent

$w = 0_v$

$\therefore W = W_1 \oplus W_2 \oplus \cdots \oplus W_r$

Let $P$ = subspace of $K$ that is not in $W$

Augment basis of $P$ to $W$ to get $[B]$ s.t $T$ has jordan form w.r.t it.

* Lemma 4.27 If $T$ is a linear operator on a finite dimensional complex vectorspace $V$.

If for some $n \in \mathbb{Z}$, $T^n = I \implies T$ is diagonalizable

# Orthogonal Linear Operator

* **Defn 5.1** $A \in \mathbb{R}^{n \times n}$ is called orthogonal if $A^T A = I$

* **Dot product** **5.2** Let $u, v \in \mathbb{R}^n$, $\underbrace{\langle u, v \rangle}_{\text{Dot Product}} := x_1 y_1 + x_2 y_2 - - + x_n y_n$ where $X = \{x_1, x_2 - -, x_n\}^t$, $Y = \{y_1, y_2 - -, y_n\}^t$ are the coordinate vectors of $u, v$ wrt the standard basis

* **Lemma 5.3** $\forall\ u, v, w \in \mathbb{R}^n$ and $a \in \mathbb{R}$

(i) $\langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle$

(ii) $\langle u, av \rangle = a \langle u, v \rangle$

(iii) $\langle u, v \rangle = \langle v, u \rangle$

* **Defn 5.4** Let $v \in \mathbb{R}^n$, then $\underbrace{\|v\|}_{\text{length of } v} := \langle v, v \rangle$

* **Defn 5.5** $u, v \in \mathbb{R}^n$ are called orthogonal if $\langle u, v \rangle = 0$

* **Defn 5.6** An basis $[\mathcal{B}] = \{v_1, v_2 - -, v_n\}$ on $\mathbb{R}^n$ is called orthonormal if

$$\forall\ 1 \le i, j \le n \qquad \langle v_i, v_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

* **Defn 5.7** A $\underbrace{T: \mathbb{R}^n \to \mathbb{R}^n}_{\text{linear operator}}$ is called orthogonal if $\forall\ u, v \in \mathbb{R}^n \qquad \langle T(u), T(v) \rangle = \langle u, v \rangle$

* **Lemma 5.8** An orthogonal linear operator $T: V \to V$ preserves the length of vectors

* **Lemma 5.9** The transformation matrix $A$ of an orthogonal linear operator $T: \mathbb{R}^n \to \mathbb{R}^n$ w.r.t the standard basis is orthogonal, ie $A^T A = I$

* **Defn 5.10** A linear operator $T: \mathbb{R}^2 \to \mathbb{R}^2$ is called a reflection if it has two eigen vector $v_1$ and $v_2$ s.t (i) $\langle v_1, v_2 \rangle = 0$ (ii) their eigen values are 1 and -1 respectively

* **Defn 5.11** $R_\theta = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix}$ $\quad S_0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\quad R_\theta S_0 = S_0 R_{-\theta} = \begin{bmatrix} c\theta & s\theta \\ s\theta & -c\theta \end{bmatrix}$

* **Lemma 5.12** Let $A$ be the transformation matrix of an orthogonal linear operator $T: \mathbb{R}^2 \to \mathbb{R}^2$ then

$A \in O_2$

$\det(A) = 1$ ↙ ↘ $\det(A) = -1$

$A = R_\theta$ $\qquad$ $A = R_\theta S_0 = S_0 R_{-\theta}$

↓ $\qquad$ ↓

Rotation about origin by $\theta$ $\qquad$ Reflection about a line making $\theta/2$ w.r.t $e_1$

Hint:- $A = \begin{bmatrix} c\theta & * \\ s\theta & * \end{bmatrix}$ $\quad A^T A = \begin{bmatrix} 1 & 0 \\ 0 & \pm 1 \end{bmatrix}$

* **Defn 5.13** An orthogonal linear operator $T: \mathbb{R}^3 \to \mathbb{R}^3$ is called a rotation if

(i) It fixes a unit vector $u$, called it's pole

(ii) It rotates the 2D subspace othogonal to $u$

* **Lemma 5.14** If $A \in SO_3 \implies 1$ is an eigen value

Hint Prove $\det(A - I) = -\det(A - I)$

* **Lemma 5.15** $\{$ Linear operators that act as rotation on $\mathbb{R}^3\}$ $\xrightarrow[\text{bijective}]{f}$ $SO_3$

* **Lemma 5.16** Let $M \in SO_3$ represents the rotation $\rho(u, \alpha)$

(i) $\text{Trace}(M) = 1 + \cos(2\alpha)$

(ii) If $B \in SO_3$ then $M' = BMB^T$ represents the rotation $\rho(u', \alpha)$ where $u' = Bu$.

Hint: (i) 4.20

(ii) to prove $\alpha' = \alpha$ use $f: \overline{B} \to \alpha'$ (rotation angle of $B \in SO_3$ → rotation angle of $M'$)

intermediate Value theorem is continuous +

* Lemma 5.17 Let $\{p_k\}_{k=1}^{\infty}$ be a sequence of monic polynomial $p_k(t)$ of degree $n$ with roots $\alpha_{k,1}, \alpha_{k,2}, \dots, \alpha_{k,n}$. Let $p(t)$ be another monic polynomial of degree $n$ with roots $\alpha_1, \alpha_2, \dots, \alpha_n$ then

(i) If $\lim_{k\to\infty} \alpha_{k,i} = \alpha_i \quad \forall\ 1 \le i \le n$ then $\lim_{k\to\infty} p_k(t) \to p(t)$

(ii) If $\lim_{k\to\infty} p_{k_i}(t) = p(t)$ then the roots of each $p_k(t)$, ie. $\alpha_{k,i}$'s can be renumbered s.t

$$\lim_{k\to\infty} \alpha_{k,i} = \alpha_i \quad \forall\ 1 \le i \le n$$

Hint (i) co-efficients of $p_k(t)$'s are continuous functions of roots

(ii) Let's renumber different roots of $p_k(t)$'s s.t $|\alpha_{k,1} - \alpha_1|$ has the smallest magnitude then

$$|\alpha_1 - \alpha_{k,1}|^n \le |\alpha_1 - \alpha_{k,1}|\,|\alpha_1 - \alpha_{k,2}| \cdots |\alpha_1 - \alpha_{k,n}| = |p_k(\alpha_1)|$$

$$\lim_{k\to\infty} \alpha_{k,1} = \alpha_1$$

$p(t) = (t-\alpha_1) q(t)$ $\quad$ $p_k(t) = (t-\alpha_{k,1}) q_k(t)$ $\quad$ show $\lim_{k\to\infty} q_k(t) = q(t)$ and use induction

* Lemma 5.18 Let $A$ be an $n \times n$ complex matrix then

(i) $\exists\ \{A_k\}_{k=1}^{\infty}$ s.t $A_k \in \mathbb{C}^{n\times n}$, has distinct roots and $\lim_{k\to\infty} A_k = A$

(ii) If $\{A_k\}_{k=1}^{\infty}$ s.t $\lim_{k\to\infty} A_k = A \implies \lim_{k\to\infty} p_k(t) = p(t)$ where $p_k(t)$ and $p(t)$ are characteristic polynomial of $A_k$ and $A$ respectively

Hint: Use Schur's lemma 4.23

* Cayley Hamilton: Theorem 5.19 Let $p(t) = t^n + c_{n-1}t^{n-1} \cdots + c_0$ be the characteristic polynomial of $A \in \mathbb{C}^{n\times n}$ then

$$p(A) = A^n + c_{n-1}A^{n-1} \cdots + c_0 I = 0_{n\times n}$$

Hint: Schur's lemma 4.23, 5.18

* Lemma 5.20 Let $\sum_{k=1}^{\infty} m_k$ be a series of +ve real numbers s.t $\sum_{k=1}^{\infty} m_k$ is convergent. If $\{u_k(t)\}_{k=1}^{\infty}$ is sequence of functions $u_k : [r,s] \to \mathbb{R}$ s.t $\|u_k(t)\|_{[r,s]} \le m_k$. Then $\sum_{k=1}^{\infty} u_k(t)$ converges uniformly to some $f : [r,s] \to \mathbb{R}$.

Proof: Let $|f_k(t)| := |\sum_{i=1}^{k} u_i|$ then $|f_k(x)| = |\sum_{j=1}^{k} u_j(x)| \le \sum_{k=1}^{\infty} m_k < M$

$\therefore$ $\{f_k\}_{k=1}^{\infty}$ is a sequence of bounded functions

Let $\varepsilon>0$, the sequence $\{S_k\}_{k=1}^{\infty}$ where $S_k := \sum_{i=1}^{k} m_i$ is convergent and therefore cauchy

$\exists\ L\in\mathbb{N}$ st $\forall p,q \geq L$ with $p\geq q$ $\quad |\sum_{i=q+1}^{p} m_i| < \varepsilon$

Consider $x\in[r,s]$ $\quad |f_p(x) - f_q(x)| = |\sum_{i=q+1}^{p} u_i(x)| \leq \sum_{i=q}^{p} m_i < \varepsilon$

$\|f_p - f_q\|_{[r,s]} < \varepsilon$

$\therefore \{f_k(x)\}_{k=1}^{\infty}$ converges uniformly to $f:[r,s]\to\mathbb{R}$

* Lemma 5.21 Let $\{u_k(t)\}_{k=1}^{\infty}$ be a sequence of continuously differentiable functions on interval $[r,s]$. Suppose $\sum_{k=1}^{\infty} u_k(t)$ converges to $f(t)$ and $\sum_{k=1}^{\infty} u_k'(t)$ converges uniformly to $g(t)$. Then $\sum_{k=1}^{\infty} u_k(t)$ converges uniformly to $f(t)$, $f(t)$ is continuously differentiable and

$f'(t) = g(t)$

Hint :- Let $f_k(t) := \sum_{i=1}^{k} u_i(t)$ $\qquad f_k'(t) = \sum_{i=1}^{k} u_i'(t)$

Then use Theorem 7.9 from real analysis

* Defn 5.20 Vector valued function, matrix valued function and their derivatives

$$x(t) := \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} \qquad x'(t) := \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ \vdots \\ x_n'(t) \end{bmatrix} \qquad A(t) := \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1,m}(t) \\ \vdots & a_{22}(t) & & \vdots \\ a_{n,1}(t) & & & a_{n,m}(t) \end{bmatrix} \qquad A'(t) := \begin{bmatrix} a_{11}'(t) & \cdots & a_{1,m}'(t) \\ \vdots & & \vdots \\ a_{n,1}'(t) & \cdots & a_{n,m}'(t) \end{bmatrix}$$

Lemma 5.21 If $A_1, A_2, \ldots, A_k$ are differentiable $n\times n$ matrices then

$$\frac{d}{dt}(A_1 A_2 \cdots A_k) = \sum_{i=1}^{k} A_1 A_2 \cdots A_{i-1} \frac{dA_i}{dt} A_{i+1} \cdots A_k$$

Defn 5.22 If $a\in\mathbb{C}$ and $a = x+iy$ then $e^a := e^{x+iy} = e^x(\cos(y) + i\sin(y))$

Defn 5.23 If $A\in\mathbb{C}^{n\times n}$

$$e^A := I + \frac{A}{1!} + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

* Lemma 5.24 (i) If $A, B \in \mathbb{C}^{n\times n}$ then $\|AB\| \leq n\|A\|.\|B\|$ where $\|A\|$ denotes maximum absolute value of $A$

(ii) If $A\in\mathbb{C}^{n\times n}$ then $\forall\ k>0$ $\quad \|A^k\| \leq n^{k-1}\|A\|$

**Theorem 5.25** (i) The series converges absolutely and uniformly for bounded sets of complex matrices.

(ii) $e^{tA}$ is differentiable function of $t$ and $\frac{d}{dt}(e^{tA}) = Ae^{tA}$

(iii) If $A, B \in \mathbb{C}^{n\times n}$ s.t $AB = BA$ then $e^{(A+B)} = e^A e^B$

Hint :- (i) Use 5.23, 5.24 and 5.20

(ii) Use 5.25(i) with 5.21

* **Lemma 5.26** $e^A$ where $A \in \mathbb{R}^{n\times n}$ is an invertible matrix

* **Lemma 5.27** If A is a jordan block of size k : $e^A = e^{\lambda I + N} = e^{\lambda I} e^N$ (commute), compute the k-1 terms as $N^k = 0$

* **Defn 5.28** Dependent variable and its derivative only appear raised to power 1 and not multiplied together; The highest derivative is first order

Linear, First Order, Constant co-efficient differential equation

$$\frac{dx}{dt} + P(y)x = Q(y) \longrightarrow (a)$$

If $Q(y) = 0$ then the equation is called homogeneous linear, first order, constant co-efficient differentiable equation.

$$\frac{dx}{dt} + P(y)x = 0 \longrightarrow (b)$$

* Every solution of (a) is called a particular solution
* Solutions of (b) are called Homogeneous solution

* **Lemma 5.29** For equation (a): General Solution = Particular solution + Homogeneous Solution

Proof :- Let $x_1(t)$ and $x_2(t)$ be two particular solutions then

$$\frac{d}{dt}(x_2(t) - x_1(t)) + P(y)(x_2(t) - x_1(t)) = Q(y) - Q(y) = 0$$

$x_2(t) - x_1(t) = x_h(t)$ } homogeneous solution

$x_2(t) = x_1(t) + x_h(t)$

* Lemma 5.30

Let $x(t)$ be a complex valued solution of $\frac{dx}{dt} = \underbrace{A}_{n\times n \text{ real matrix}} x$ $\Longrightarrow$ The real and complex part of $x(t)$ solve the same equation

Proof Let $x(t) = u(t) + iv(t)$

$$\frac{d}{dt}(u+iv) = Au + iAv$$

Comparing real and complex parts

$$\frac{du}{dt} = Au \qquad \frac{dv}{dt} = Av$$

* Lemma 5.31

Let $A \in \mathbb{R}^{n\times n}$ and let $P$ be an invertible matrix s.t $A' = P^{-1}AP = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix}$

Then the general solution of $\frac{dx}{dt} = Ax$ is $x = P\tilde{x}$ where

$\tilde{x} = (c_1 e^{\lambda_1 t}, c_2 e^{\lambda_2 t}, \dots, c_n e^{\lambda_n t})$ and solves $\frac{d\tilde{x}}{dt} = A'\tilde{x}$

* Theorem 5.32

Let $A \in \mathbb{R}^{n\times n}$ or $\mathbb{C}^{n\times n}$. The columns of matrix $e^{tA}$ form a basis for the solution space of $\frac{dx}{dt} = Ax$

Proof $\frac{d}{dt}\left(e^{-tA}x(t)\right) = -Ae^{-tA}x(t) + e^{-tA}Ax = 0$

$\therefore e^{-tA}x(t) = c$

$x(t) = e^{tA}c$

# Symmetry

* **Defn 6.1** Distance b/w $P, Q \in \mathbb{R}^n$ (vector pointing to Q from origin, vector pointing to P from origin)

$$|\vec{PQ}| := |v - u|$$

(Length of vector $5.4$)

* **Defn 6.2** Isometry (Distance preserving map): $f: \mathbb{R}^n \to \mathbb{R}^n$ s.t $\forall\, u, v \in \mathbb{R}^n$

$$|f(u) - f(v)| = |u - v|$$

* **Lemma 6.3** Let $T: \mathbb{R}^n \to \mathbb{R}^n$ be a map, then

T is an isometry that fixes the origin $\iff$ T is an orthogonal linear operator

Hint: if $\langle x, y\rangle = \langle x, x\rangle = \langle y, y\rangle$ then $x = y$

* **Lemma 6.4** All the isometries of $\mathbb{R}^n$ together form a group denoted by $M_n$

* **Lemma 6.5** If $m \in M_n$, then $\exists$ a unique translation $t_a \in M_n$ and an orthogonal linear operator $\varphi \in M_n$ s.t

$$m = t_a \varphi$$

Hint 6.3

* **Lemma 6.6** Let $t_a, \varphi \in M_n$ be an translation and orthogonal linear operator respectively then

$$\varphi t_a = t_{\varphi(a)} \varphi$$

* **Lemma 6.7** (i) $\pi: M_n \to O_n$ defined by $\pi(t_a \varphi) := \varphi$ is a group homomorphism

(ii) $m \in M_n$: $\det(\pi(m)) = 1$ → Orientation preserving isometry; $\det(\pi(m)) = -1$ → Orientation reversing isometry

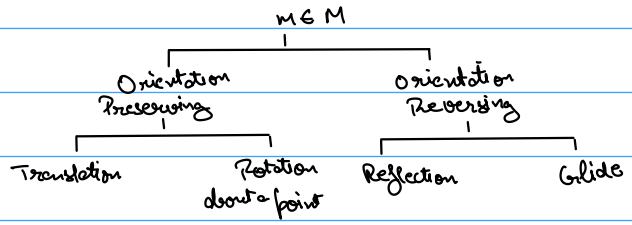
* **Lemma 6.8** Let $C_1, C_2$ be two coordinate systems (Origin + Orthonormal Basis) of $\mathbb{R}^n$ associated with an isometry $\eta \in M_n$, $T \in M_n$ s.t its formula w.r.t $C_1$ and $C_2$ are $m_1$ and $m_2$ respectively then ($P_{C_1} = \eta P_{C_2}$)

$$m_2 = \eta^{-1} m_1 \eta$$

* **Defn 6.9** $M$: Isometry group of $\mathbb{R}^2$.

↳ generated by: $t_a$ (Translation by vector $a$), $S_\theta$ (rotation by angle $\theta$ about origin), $r$ (reflection about $e_1$ axis)

* Lemma 6.10




* Lemma 6.11 — If $G \subset M$ s.t it preserves a point $p \Longrightarrow$ $G \subset t_p O_2 t_p^{-1}$

* Defn 6.12 — $\Gamma \in (\mathbb{R}, +)$ is called discrete if $\exists\, \varepsilon > 0$ s.t $\underbrace{|a|}_{\text{absolute value}} > \varepsilon \quad \forall a \in \Gamma$

* Lemma 6.13 — If $\Gamma \in (\mathbb{R}, +)$ is discrete $\Longrightarrow$ $\Gamma = a\mathbb{Z}$ where $a \in \mathbb{R}$

* Defn 6.14 — $G \subset M$ is called discrete if $\exists \{m_k\}_{k=1}^{\infty}$ where $\forall k \in \mathbb{N}$ $m_k \in G$, $\forall j,k \in \mathbb{N}$ s.t $j \neq k$ $m_j \neq m_k$ and $\lim_{k\to\infty} m_k = I$

* Lemma 6.15 — $G \subset M$ is discrete $\Longleftrightarrow$ $\exists\, \varepsilon > 0$ s.t
- $\forall\, t_a \in G$ with $a \neq 0$ then $\underbrace{|a|}_{\text{length of vector}} > \varepsilon$
- $\forall\, t_a S_\theta \in G$, $\theta \neq 0$ then $|a| > \varepsilon$

Hint: Use contradiction, Case 1 finite number of orientation reversing Case 2 infinite number of orientation reversing elements, $\det$ is a continuous function of elements of matrices.

* Lemma 6.16 — Let $G \subset O_2$, then $G$ has finite order $\Longleftrightarrow$ $G$ is discrete

- $C_n$: Cyclic group generated by $S_\theta$ where $\theta = \frac{2\pi}{n}$ and $n \in \mathbb{N}$
- $D_n$: Dihedral group. if coordinates system is selected appropriately generated by $r$ and $S_\theta$ and $\theta = \frac{2\pi}{n}$ and $n \in \mathbb{N}$

* Fixed Point Theorems 6.17 — If $G \subset M$ has finite order $\Longrightarrow$ $G$ fixes a point and $G = C_n$ or $D_n$

* **Defn 6.18** The translation group $G \subset M$ is defined as

$$L := \{ v \in \mathbb{R}^2 \text{ s.t } t_v \in G \}$$

* **Lemma 6.19** Let $L \subset (\mathbb{R}^2, +)$ be discrete then

(i) A bounded region of $(\mathbb{R}^2, +)$ contains only finitely many point of $L$

(ii) If $L$ is non trivial then it contains a non zero vector of minimal length.

* **Lemma 6.20** Let $B = \{u, v\}$ be a basis of $(\mathbb{R}^2, +)$, $L := \{ mu + nv \text{ where } m, n \in \mathbb{Z} \}$

$$\pi'(B) := \{ p \in (\mathbb{R}^2, +) \text{ s.t either } p \text{ is inside the } \parallel\text{gram with vertices } \{0, u, v, u+v\} \text{ or } p \text{ lies along the edges } [0,u) \text{ and } [0,v) \}$$

then $\forall w \in (\mathbb{R}^2, +)$ $\exists\, x \in L$ and $w_0 \in \pi'(B)$ st $w = x + w_0$.

Hint division by remainder

* **Lemma 6.21** If $L \subset (\mathbb{R}^2, +)$ is discrete then

- $L = \{0\}$
- $L = \{ a\mathbb{Z} \text{ where } a \in \mathbb{R}^2 \}$
- $L = \{ a\mathbb{Z} + b\mathbb{Z} \text{ where } a, b \in \mathbb{R}^2 \}$

* **Defn 6.22** The point group $\bar{G}$ of $G \subset M$ is defined as

$$\bar{G} := \{ \varphi \in O_2 \text{ s.t } t_a \varphi \in G \}$$

* **Theorem 6.23** Let $G \subset M$ be discrete, $a \in L$, $\bar{g} \in \bar{G}$ and $a' = \bar{g}(a) \Longrightarrow a' \in L$

Hint : Use conjugation

* **Lemma 6.24** Crystallographic restriction :- Let $L$ be a non trivial discrete subgroup of $(\mathbb{R}^2, +)$, $H \subset O_2$ be the subgroup of symmetries of $L$, then

* Every rotation in $H$ has one of these order $1, 2, 3, 4, 6$
* $H$ has the structure of $C_n$ or $D_n$ with $n = 1, 2, 3, 4, \text{or } 6$

* Note 6.25 The point group $\bar{G}$ and the translation group $L$ of a discrete group $G$ doesn't identify it completely.

* Lemma 6.26 The concept of crystallographic restrictions limits the number of discrete subgroups of $M$ to 17.

* Note 6.27 The concept of crystallographic restrictions can be extended to 3D to conclude there are 230 discrete groups in 3D.

* Defn 6.28 Let $G$ be a group and $S$ be a set. Then the map $f: G \times S \to S$ is called a group operation if

(i) $e.s = s \quad \forall s \in S$

(ii) $g_1.(g_2.s) = (g_1.g_2).s \quad \forall s \in S, \forall g_1, g_2 \in G$

* Defn 6.29 Let $f: G \times S \to S$ be a group operation, $s \in S$

Orbit of $s$ $\quad O_s := \{ g.s : g \in G \}$

Stabilizer of $s$ $\quad G_s := \{ g \in G : g.s = s \}$

* Defn 6.30 A group operation $f: G \times S \to S$ is called transitive if $S$ consists of a single orbit under $f$.

* Example Let $H \subset G$, then $f: G \times G/H \to G/H$ defined by $f(g, \bar{a}) = \overline{ga}$ is a transitive group action

* Lemma 6.31 Let $f: G \times S \to S$ a group operation, $s \in S$, $H \subset S$ be the stabilizer of $s$ then

(1) If $a, b \in G$ s.t $as = bs \implies a^{-1}b \in H$

(2) If $s' \in S$ s.t $s' = g.s$ then the stabilizer of $s'$, ie $H' = gHg^{-1}$

* Theorem 6.32 Let $f: G \times S \to S$ a group operation, $s \in S$, $O_s$ and $H \subset S$ be the orbit and stabilizer of $s$. Then $\exists$ a bijective map $\varepsilon: G/H \to O_s$ where $\varepsilon([aH]) := a.s$.

Also $\quad |G| = |H|\,|G/H|$ $\quad$ } Counting Formula

$\quad\quad = |H|\,|O_s|$

* Lemma 6.33 — The relation of belonging to an orbit is an equivalence relation $\Rightarrow$ The orbits partition the set $S$ & $|S| = |O_{s,1}| + |O_{s_2}| + \cdots + |O_{s_k}|$

* Defn 6.34 — The permutation representation of $f: G \times S \to S$ a group operation is a group homomorphism $\varphi: G \to \text{Perm}(S)$
($\varphi$: permutation representation; Perm(S): Group of permutations of $S$)

* Lemma 6.35 — $\{ f: G \times S \to S \}$ (Operation of $G$ on set $S$) $\xleftrightarrow{\text{bijective correspondence}}$ $\{ \varphi: G \times S \to S \}$ (permutations representations)

* Defn 6.36 — A group operation $f: G \times S \to S$ is called faithful if the only element $g \in G$ s.t. $g.a = a$ $\forall a \in S$ is identity, i.e. $e \in G$.
Restatement: The kernel of the permutation representation is $\{e\}$

* Theorem 6.37 — Let $G \subset SO_3$ be a finite ordered subgroup. Let $P$ be the set of poles of elements of $G$. Then $f: G \times P \to P$ is a valid group operation where
$f(g,p) := g(p)$

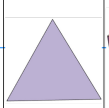
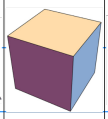
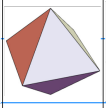
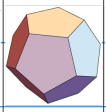
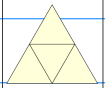
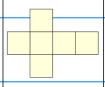
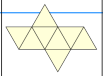
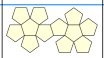
* Lemma 6.38 — A finite subgroup of $SO_3$ has one of the following structure
(i) $C_k$ :- Cyclic of order $k$
(ii) $D_k$ :- Dihedral group of order $2k$.
(iii) $T$ :- Tetrahedral group of 12 rotational symmetries
(iv) $O$ :- Octahedral group of 24 rotational symmetries
(v) $I$ :- Icosahedral group of 60 rotational symmetries

Hint (a) Use 6.37 & count the number of fixes to get
$\sum_{O_p} \left(1 - \frac{1}{r_p}\right) = 2 - \frac{2}{N}$
($O_p$: Orbit of pole $p$; $\#$ of elements with $p \in P$ as pole; $N$: Order of group)
Also $N = r_p \, n_p$ ($n_p$: # elements in the orbit of $p \in P$)

(b) Use $1 \leq 2 \cdot 2/N < 2$

(c) $1 + 2/N = 1/q_1 + 1/q_2 + 1/q_3$ ∴ It is not possible $q_1 > 3$, $q_2 > 3$ and $q_3 > 3$

* Lemma 6.39 There exist only 5 regular polyhedron

| regular tetrahedron | cube | regular octahedron | regular dodecahedron | regular icosahedron |
|---|---|---|---|---|
|  |  |  |  |  |
|  |  |  |  |  |

Symmetry Groups: T O O I I

# $p$ Groups



* **Defn 7.1** Let $G$ be a group. Then $f: G \times G \to G$ defined by $f(g,x) := gxg^{-1}$ is called conjugation and is a valid group operation

* **Defn 7.2** Let $G$ be a group. Then

(i) $Z := \{x \in G : x.g = g.x \ \forall g \in G\}$ is called center of $G$

(ii) $Z(x) := \{g \in G : g.x = xg\}$ is called Centralizer of $x$

* **Lemma 7.3** Let $G$ be a group and $x \in G \implies Z \subset Z(x)$, $x \in Z(x)$

* **Lemma 7.4** (i) $|G| = |Z(x)| \, |C(x)|$ Hint 6.32

$C(x)$: Conjugacy class of $x$

(ii) $|G| = |C(x_1)| + |C(x_2)| - \cdots + |C(x_k)|$

* **Lemma 7.5** (i) If $x \in Z \implies C(x) = \{x\}$

(ii) If $x \in H$ and $H \trianglelefteq G \implies C(x) \subset H$

* **Defn 7.6** A group $G$ is called $p$ group if $|G| = p^e$ where $p$ is a prime number and $e \geq 1$

* **Lemma 7.7** The center $Z$ of a $p$ group is non trivial. Hint 7.5 (ii)

* **Lemma 7.8** Let $G$ be a $p$ group, $S$ be a set s.t $|S| = n$ and $n$ is not divisible by $p$, $\mathcal{S}: G \times S \to S$ be a valid group operation, then $\exists\, s \in S$ s.t $|O_s| = 1$.

Hint 7.5 6.32

* **Lemma 7.9** Every group $G$ of order $p^2$ where $p$ is a prime number is :

(i) Abelian. Hint 7.4

(ii) Either cyclic or isomorphic to the product of two cyclic groups of order $p$.

* **Defn 7.10** A group $G$ is called simple if it has no proper normal group

* **Defn 7.11** Let $G$ be a group and $S := \{ H \subset G \}$ (set of subgroups of $G$) then

$f: G \times S \to S$ defined by $f(g,H) := gHg^{-1}$ is a valid group operation

$N(H) = \{ g \in G : gHg^{-1} = H \}$

stabilizer of $H$

* **Lemma 7.12** Let $G$ be a group and $H \subset G$ then

$H \trianglelefteq N(H) \subset G$

$H \trianglelefteq G \iff N(H) = G$

* **Lemma 7.13** Let $S$ be a set s.t $|S| = p^e m$ where $p$ is a prime number, $e \geq 1$, and $m$ is not divisible by $p$

$U := \{ A \subset S \text{ s.t } |A| = p^e \}$

then $|U| = {}^{p^e m}C_{p^e}$ is not divisible by $p$

* **Lemma 7.14** If $G$ is a group, $S := \{ A \subset G \}$ (set of subset of $G$) then

(i) $f: G \times S \to S$ where $f(g,A) := gA$ } is a valid group operation.

(ii) if $U \subseteq S$, $H \subset G$ is stabilizer of $U$ then $|H|$ divides $|G|$ and $|U|$

Hint: $H \times U \to U$ (right cosets)

$|U| = |O_{u_1}| + |O_{u_2}| + \cdots + |O_{u_k}|$

* **Defn 7.15** Let $G$ be a group s.t $|G| = p^e m$ where $p$ is a prime number, $e \geq 1$, and $m$ is not divisible by $p$.

$H \subset G$ is called sylow's $p$ group if $|H| = p^e$.

* **Sylow's First Theorem 7.16** Let $G$ be a group s.t $|G| = p^e m$ where $p$ is a prime number, $e \geq 1$, and $m$ is not divisible by $p$.

Then $\exists\, H \subset G$ s.t $H$ is a sylow's $p$ group

Hint $f: G \times S \to S$ (Set of subset of $G$ of order $p^e$)

Use 7.13, 7.8

$|S| = |O_{s_1}| + |O_{s_2}| + \cdots + |O_{s_k}|$

* Second Sylow's Theorem 7.17

Let $G$ be a group s.t $|G| = p^e m$ where $p$ is a prime number, $e \geq 1$, and $m$ is not divisible by $p$

(i) If $K \subset G$ is a $p$ subgroup and $\Rightarrow$ $\exists g \in G$ s.t $K \subset gHg^{-1}$
$H \subset G$ a sylow's $p$ group

(ii) All sylow's $p$ subgroups are conjugates of each other.

Hint $f: G \times G/H \to G/H$ where $f(g,[x]) = g[x]$ is a transitive group operation (using 6.30)
Stabilizer of $[H]$ is $H$ itself.

consider $f': K \times G/H \to G/H$

then $|G/H| = m$ is not divisible by $|K|$ using 7.8

$\exists [L] \in G/H$ s.t $[L]$ is fixed by $K \Rightarrow K \subset M_{[L]}$ where $M_{[L]} \subset G$ is stabilizer of $[L]$

also $\exists g \in G$ s.t $g[H] = [L] \Rightarrow M_{[L]} = gHg^{-1}$ using 6.31

$K \subset gHg^{-1}$

* Sylow's Third Theorem 7.18

Let $G$ be a group s.t $|G| = p^e m$ where $p$ is a prime number, $e \geq 1$, and $m$ is not divisible by $p$

If $s$ is the number of sylow's $p$ subgroup, then

(i) $m$ is divisible by $s$

(ii) $s = kp+1$

Hint $f: G \times \overbrace{S}^{\text{set of subgroups } p \text{ subgroup}} \to S$ defined by $f(g,H) := gHg^{-1}$ is a transitive group operation (7.17)

let $H \in S$ then $|G| = |N(H)|\,|O_H|$

$= |N(H)|\,|S|$ and $H \subset N(H) \Rightarrow p^e \leq N(H) \leq p^e m$

$p^e m = p^e a s$ where $e \in \mathbb{N}$

$\therefore$ $s$ divides $m$.

(ii) $f': H \times S \to S$ then $|O_H| = 1$

let $H' \in S$ s.t $|O_{H'}| = 1$ then $H \subset N(H') \subset G$

$H', H$ are sylow's $p$ subgroup of $N(H')$ where $|N(H')| = p^e a$ $\therefore$ $\exists g \in N(H')$

s.t $gH'g^{-1} = H$ since $H' \trianglelefteq N(H') \Rightarrow H = H'$ $\therefore$ only $H$ has orbit 1